

Assignment - 1Section - A

1. What is a DBMS? Why is it required?

A DBMS is a software that manages data in an organised way using tables. It helps user to store update & retrieve data easily. It is required to manage large data safely, avoid duplication & provide secure access.

2. Why DBMS is better than file system?

DBMS stores data in structured format & allow multiple users to access it safely. File system may create duplicate data & are harder to manage.

3. What is data abstraction? Explain its levels.

It means hiding technical detail & showing only required information to users.

→ Physical - how data is stored

→ Logical - what data is stored

→ View - what user sees

4. What is three schema architecture?

It divides database into three parts

i) Internal level

ii) Conceptual level

iii) External level

It helps in data independence & easy modification.

5. Data Independence - It means changing dataset structure at one level without affecting other levels

It helps in flexibility & system maintenance

6. What is an ER model?

It is a diagram that represents entities, their attributes & relationships. It is used for database design planning.



## 7. What are Integrity constraints ?

Integrity constraints are rules that ensure correct & valid data

Ex: Primary keys, foreign key

## Section - B

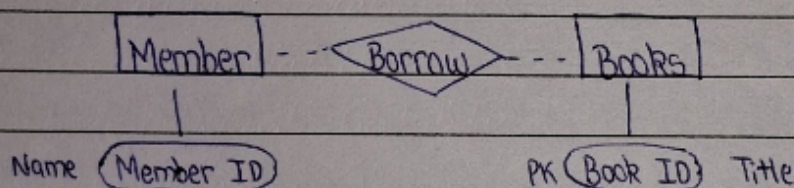
## 1. Three - schema

External level - user view

Conceptual level - overall structure

Internal level - physical storage

## 2. ER diagram → Library



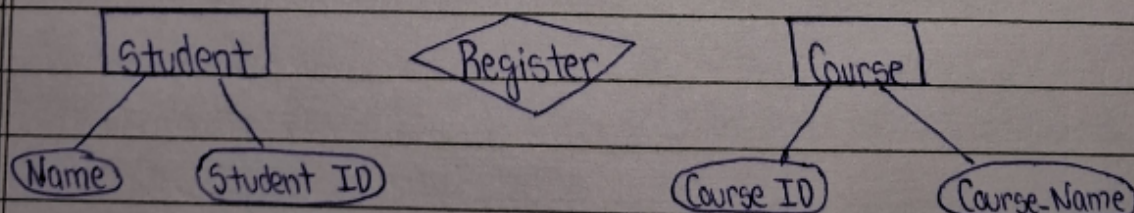
- Entities :- Member, book

- Relation :- Borrow

- PK :- Member ID, book ID

One member can borrow many books

## 3. ER → student course registration



- Students (Student ID, Name)

- Course (Course ID, Course Name)

Relation : Student register for course



## Section - C

## 1) Hospital system

Entities : Patients , Doctor , Appointment

Patients books appointment

Doctor attends appointment

Primary key : Patient ID, Doctor ID, Appointment ID

## 2) Online shopping

Customer (Customer ID - PK)

Product (Product ID - PK)

Order (Order ID - PK)

## 3) University data abstraction

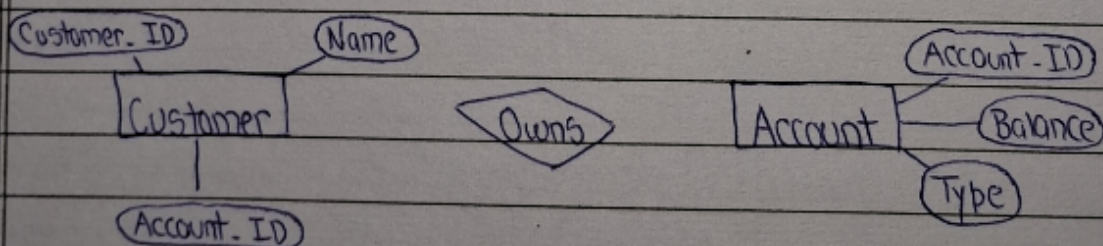
Students see marks (view level)

Admin manage records (logical level)

DBA control storage (Physical level)

## Section - D

## 1. ER diagram → ATM



Primary key is account no

Relation : Customer owns account

Account no take uniquely identified account



## 2. Three - schema architecture

Benefit : Provide data independence

External level

↓  
Conceptual level

↓  
Internal level

## 3. Draw entity with attributes

Entity : Employee

Employee ID (PK) = Primary Key

Name

Salary

Department

~~Section - C~~